

TRANSPORT PHENOMENA

Prof. Sunando DasGupta (3 classes)
and
Prof. Manish Kaushal (1 class)

TRANSPORT PHENOMENA

Basic Overview

Over the last few decades, the subject has revolutionized the way Chemical Engineering science is taught. This course deals with the unified treatment of the different transport processes, ubiquitous in industry as well as in nature. Momentum, heat and mass transfer are taught together due to the underlying similarities of the mathematics and molecular mechanisms describing such processes.

Books

1. **Transport Phenomena by Bird, Stewart and Lightfoot (Wiley)**
2. Transport Phenomena Fundamentals by J. L. Plawsky (CRC Press)
3. Heat and Mass Transfer – A Transport Phenomena Approach by K. S. Gandhi (New Age)
4. **Fluid Mechanics by R. W. Fox, P. J. Pritchard and A. T. McDonald (Wiley)**
5. **Introduction to Heat and Mass Transfer by F. P. Incropera & D. P. DeWitt (Wiley)**

Grading: 2-3 class tests for TA, relative grading

Course Plan

Formulation and solution of momentum transfer in laminar flow.

Navier-Stokes equation and its applications

Boundary Layer concepts, boundary layer thicknesses (disturbance, displacement and momentum), Blasius solution for flow over a flat plate

Use of momentum integral equation, turbulent boundary layers, fluid flow about immersed bodies, drag

Formulation and solution of heat transfer in laminar flow

Development and use of energy equation

Transient conduction - lumped capacitance, analytical solutions and other methods.

Formulation and solution of mass transfer in laminar flow. Development and use of species balance equation

Introduction to convective flow, natural convection, relevant examples from heat and mass transfer

Mathematical treatment of the similarities between heat, mass and momentum transfer, similarity parameters, and relevant analogies.

Solution of coupled heat, mass and momentum transfer problems based on analogy.

Phenomenological Relations

Governing Equations

Re

Pr

Sc

f

Nu

Sh

$f = f(\text{Re})$

$\text{Nu} = f(\text{???})$

$\text{Sh} = f(\text{???})$

TRANSPORT PHENOMENA

Application of Navier Stokes Equations

Continuity Equation

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

Equation of Motion – Navier Stokes equation

$$\rho D\bar{v} / Dt = -\nabla p - [\nabla \cdot \bar{\tau}] + \rho g$$

Cartesian Coordinate – z component

$$\begin{aligned} & \rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) \\ &= - \frac{\partial P}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \end{aligned}$$

Relevant Boundary Conditions

No slip at the liquid-solid interface

No shear at the liquid-vapor interface

The Equation of Continuity

$$[\partial\rho/\partial t + (\nabla \cdot \rho\mathbf{v}) = 0]$$

Cartesian coordinates (x, y, z):

$$\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0 \quad (\text{B.4-1})$$

Cylindrical coordinates (r, θ , z):

$$\frac{\partial\rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0 \quad (\text{B.4-2})$$

Spherical coordinates (r, θ , ϕ):

$$\frac{\partial\rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r}(\rho r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(\rho v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}(\rho v_\phi) = 0 \quad (\text{B.4-3})$$

^a When the fluid is assumed to have constant mass density ρ , the equation simplifies to $(\nabla \cdot \mathbf{v}) = 0$.

The Equation of Motion for a Newtonian Fluid with constant μ , ρ

The Navier Stokes Equation

Cartesian coordinates (x, y, z):

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left[\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right] + \rho g_x$$

$$\rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left[\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right] + \rho g_y$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

Cylindrical coordinates (r, θ, z) :

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) &= -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r \\ \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta \\ \rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \end{aligned}$$

Spherical coordinates (r, θ, ϕ) :

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\theta^2 + v_\phi^2}{r} \right) &= -\frac{\partial p}{\partial r} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial^2}{\partial r^2} (r^2 v_r) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_r}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_r}{\partial \phi^2} \right] + \rho g_r \\ \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} + \frac{v_r v_\theta - v_\phi^2 \cot \theta}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\theta}{\partial \phi^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} - \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right] + \rho g_\theta \\ \rho \left(\frac{\partial v_\phi}{\partial t} + v_r \frac{\partial v_\phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\phi}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi v_r + v_\theta v_\phi \cot \theta}{r} \right) &= -\frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\phi}{\partial \phi^2} + \frac{2}{r^2 \sin \theta} \frac{\partial v_r}{\partial \phi} + \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right] + \rho g_\phi \end{aligned}$$

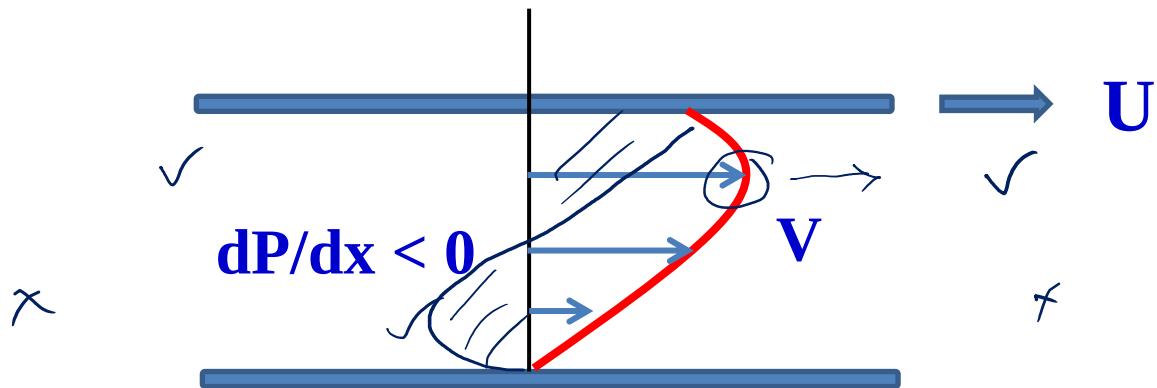
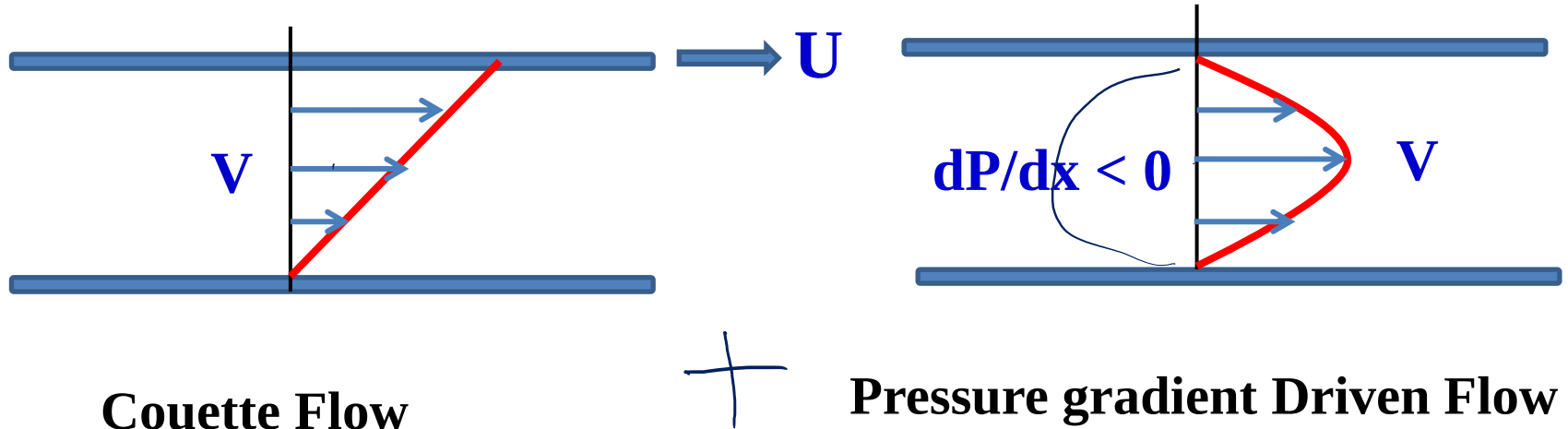
Transport Phenomena - Momentum Transfer

The record-write head for a computer disk storage system floats above the spinning disk on a very thin film of air (the film thickness is 0.5micron). The head location is 150mm from the disk centre line, the disk spins at 3600rpm. The record-write head is 10mm x 10mm square. Determine for standard air (density= 1.23kg/m^3 , viscosity= $1.78 \times 10^{-5} \text{ kg/(m.s)}$) in the gap between the head and the disk -

(a) the Reynold's number of the flow, (b) the viscous shear stress and (c) the power required to overcome viscous shear stress.

For such a small gap the flow can be considered as flow between parallel plates.

Flow between parallel plates



Couette and Pressure gradient Driven Flow

Solution

Since the gap is too small, the situation can be approximated by the case of flow between two parallel plates with one plate moving and the other stationary with no applied pressure gradient.

Here,

$$V = R\omega = 0.15 \text{ m} \times 3600 \text{ rev/min} \times 2\pi \text{ rad/rev} \times 1/60 \text{ min/s} = 56.5 \text{ m/s}$$

$$\text{Re} = (\rho V a)/\mu = (V a)/(\mu/\rho) = 56.5 \text{ m/s} \times 0.5 \times 10^{-6} \text{ m} / (1.45 \times 10^{-5} \text{ m}^2/\text{s}) = 1.95 \text{ (Laminar Flow)}$$

For Couette flow between two parallel plates, with no applied pressure gradient, the velocity profile will be linear with no-slip boundary conditions at the bottom (stationary) and top (moving) plates.

Therefore, the shear stress, τ , can be expressed as

$$\tau = \mu du/dy = \mu V/a \text{ (linear velocity profile)}$$

$$\tau = 1.78 \times 10^{-5} \text{ kg / (m.s) } \times 56.5 \text{ m/s } \times 1/(0.5 \times 10^{-6} \text{ m}) = 2.01 \text{ KN/m}^2$$

Therefore,

$$\text{Force, } F = \tau A = \tau \times (W \times L), \text{ Torque, } T = F \times R = \tau \times (WL) \times R$$

$$\text{Power dissipation rate. } P = T \times \omega = \tau WL R \omega$$

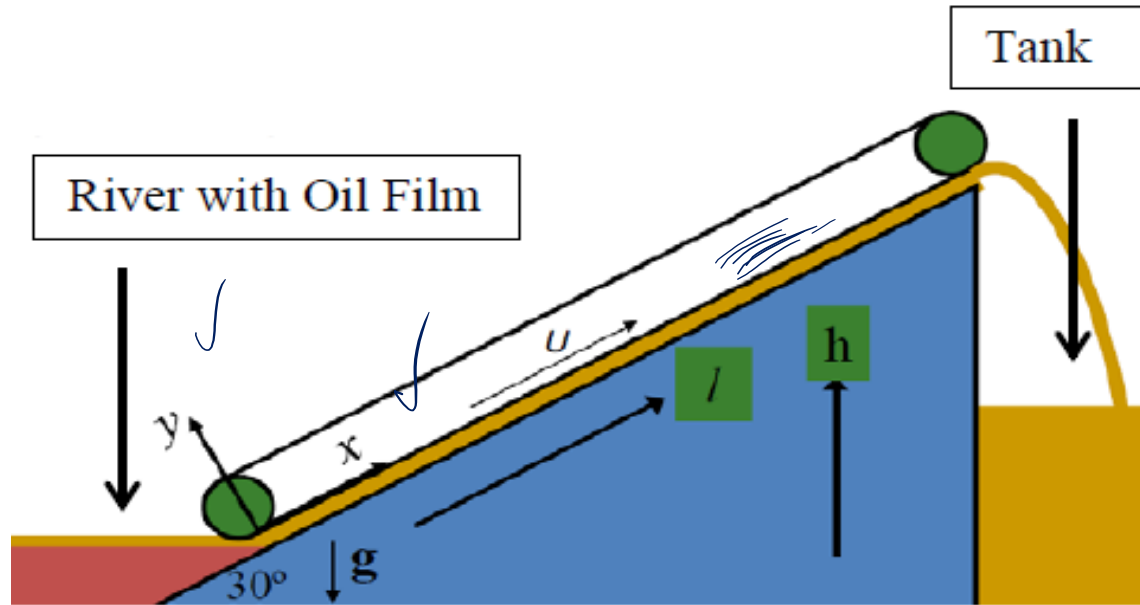
$$P = 2.01 \times 10^{-3} \times 0.01 \times 0.01 \times 0.15 \times 3600 \times 2\pi \times 1/60 = 11.4 \text{ W}$$

Thus the power required to overcome viscous shear stress is 11.4 W

Transport Phenomena - Momentum Transfer

Problem 2

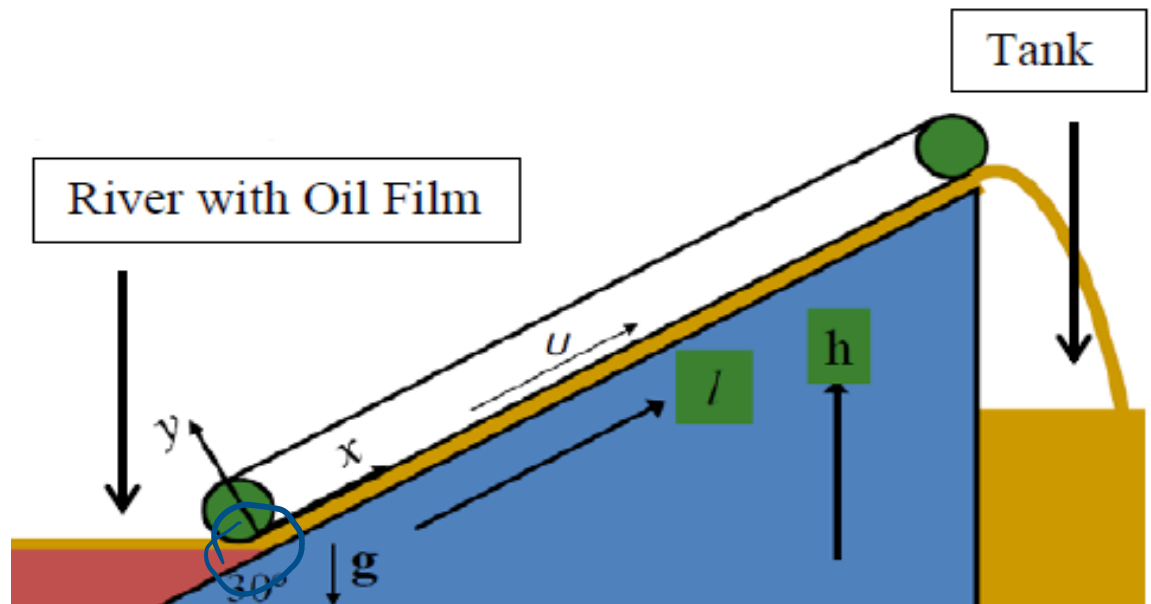
An oil skimmer uses a 5 m wide x 6 m long moving belt above a fixed platform ($\theta = 30^\circ$) to skim oil off of rivers ($T = 10^\circ\text{C}$). The belt travels at 3 m/s. The distance between the belt and the fixed platform is 2 mm.



The belt discharges into an open tank on the ship. The fluid is actually a mixture of oil and water. To simplify the analysis, assume crude oil dominates. Find the discharge of oil into the tank on the ship, the force acting on the belt and the power required (kW) to move the belt. For oil: $\rho = 860 \text{ kg/m}^3$, viscosity, $\mu = 1 \times 10^{-2} \text{ N.s/m}^2$

The x-component of the equation of motion is given as

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = - \frac{\partial P}{\partial x} + \mu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right] + \rho g_x$$



Considering one-dimensional flow ($v = 0 = w$), no applied pressure gradient, u is a function of only y , and a steady state process, the above equation reduces to the following governing equation

$$\frac{d^2u}{dy^2} = \frac{\rho g}{\mu} \sin\theta$$

The relevant boundary conditions (no slip) are

$$\text{BC 1 At } y = 0 \quad u = 0$$

$$\text{BC 2 At } y = h, \quad u = U$$

Therefore,

$$\frac{du}{dy} = \frac{\rho g \sin \theta}{\mu} y + A \quad \text{and} \quad u = \left(\frac{\rho g \sin \theta}{\mu} \right) \frac{y^2}{2} + Ay + B$$

Applying the BC s $\rightarrow$ $B = 0$ and $A = \left(\frac{\rho g \sin \theta}{\mu} \right) \frac{h}{2} + \frac{u}{h}$

Thus

$$u = - \left(\frac{\rho g \sin \theta}{\mu} \right) \left(\frac{hy}{2} - \frac{y^2}{2} \right) + \frac{Uy}{h}$$

The volumetric flow rate per unit width of the film is given by

$$Q = \int_0^h u dy = - \int_0^h \left[\left(\frac{\rho g \sin \theta}{\mu} \right) \left(\frac{hy}{2} - \frac{y^2}{2} \right) + \frac{Uy}{h} \right] dy = - \frac{\rho g h^3}{12\mu} \sin \theta + \frac{Uh}{2}$$

$$Q = - \frac{860 \times 9,81 \times 0.002}{12 \times 10^{-2}} \sin 30 + \frac{3 \times 0.002}{2} = 0.0027 \frac{m^2}{s} \text{ (per unit width)}$$

For the width of 5 m, the volumetric flow rate will be equal to 0.0135 m³/s.

Evaluate $\tau = \mu \, du/dy$ at the moving belt from the expression for velocity as

$$\frac{du}{dy} = - \left(\frac{\rho g \sin \theta}{\mu} \right) \left(\frac{h}{2} - y \right) + \frac{U}{h}$$

And at the moving belt,

$$\tau = \mu \left(\frac{du}{dy} \right)_{y=h} = \left(\frac{\rho g \sin \theta}{2} \right) h + \frac{\mu U}{h}$$



Using the values $h = 0.002\text{m}$, $\mu = 10^{-2}\text{N.s/m}^2$ and $U = 3 \text{ m/s}$

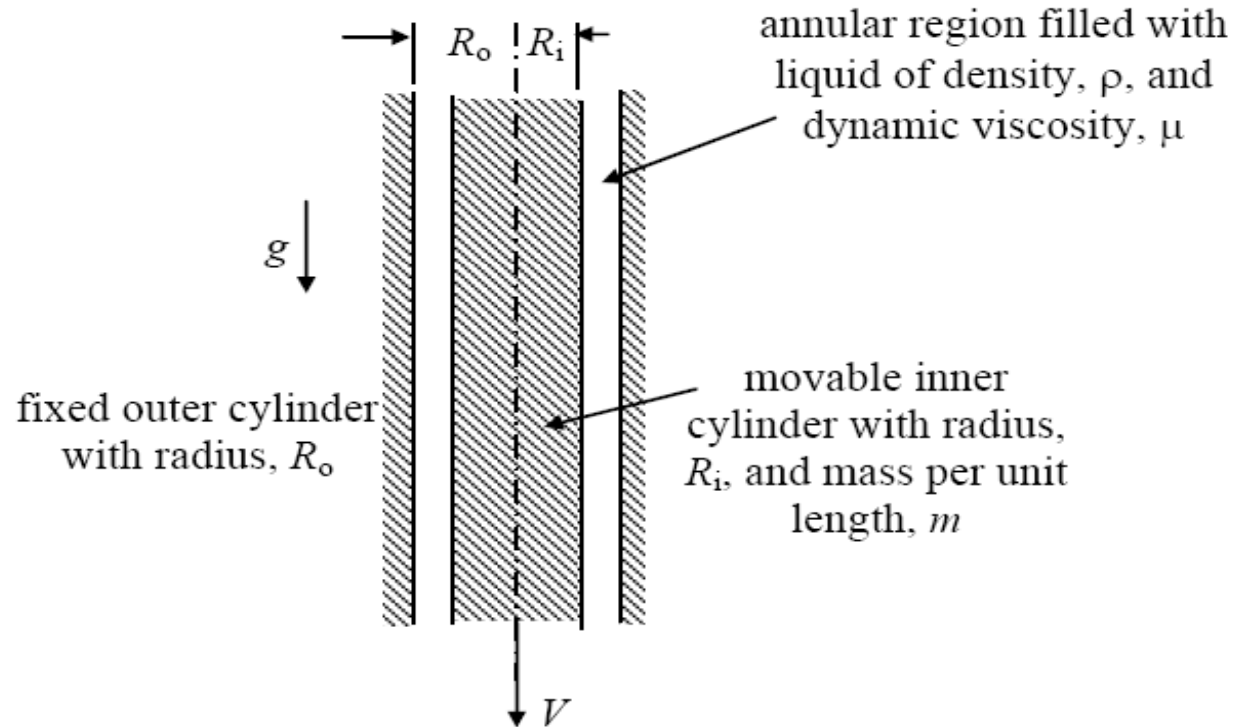
$$\tau \Big|_{\text{at the belt}} = 19.21 \text{ N/m}^2$$

$$\text{Power} = (\tau \times L \times W) U = 19.21 \times 6 \times 5 \times 3 = 1.73 \text{ KW}$$

Transport Phenomena - Momentum Transfer

Problem 3

Consider two concentric cylinders with a Newtonian liquid of constant density, ρ , and viscosity, μ , contained between them. The outer pipe, with radius, R_o , is fixed while the inner pipe, with radius, R_i , and mass per unit length, m , falls under the action of



gravity at a constant speed. There is no pressure gradient within the flow and no swirl velocity component. Determine the vertical speed, V , of the inner cylinder as a function of the following parameters: g , R_o , R_i , m , ρ , and μ . The space between the two cylinders is **not** 'too small' compared to the radii of the cylinders

Solution

For the inner cylinder moving at constant velocity, the downward force is exactly balanced by the viscous force as

$$(\tau_w A_w) \big|_{Inner\ Cylinder} = m L g$$

The z component of the equation of motion in cylindrical coordinate is

$$\begin{aligned} & \rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) \\ &= -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \end{aligned}$$

Cancelling the terms with the observations i) steady state, ii) v_z is a function of r only, not of z or θ , iii) no applied pressure gradient

The following simplified form of the NS equation can be obtained.

Governing equation

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dv_z}{dr} \right) = - \frac{\rho g}{\mu}$$

Upon integration (with C_1 and C_2 being the constants of integration)

$$v_z = - \frac{\rho g r^2}{4\mu} + C_1 \ln r + C_2$$

Boundary Conditions

$$v_z = V \text{ at } r = R_i \quad \Rightarrow \quad V = - \frac{\rho g}{4\mu} R_i^2 + C_1 \ln R_i + C_2$$

$$v_z = 0 \text{ at } r = R_o \quad \Rightarrow \quad 0 = - \frac{\rho g}{4\mu} R_o^2 + C_1 \ln R_o + C_2$$

Therefore
$$V = \frac{\rho g}{4\mu} (R_o^2 - R_i^2) + C_1 \ln \frac{R_i}{R_o}$$

$$C_1 = \frac{1}{\ln \frac{R_i}{R_o}} \left[V - \frac{\rho g}{4\mu} (R_o^2 - R_i^2) \right]$$

$$\frac{dv_z}{dr} = -\frac{\rho g r}{2\mu} + \frac{C_1}{r} \quad \tau_{rz} = \mu \frac{dv_z}{dr} = -\frac{\rho g r}{2} + \frac{C_1 \mu}{r}$$

Since the force on the inner cylinder = force due to gravity

$$\tau \big|_{r=R_i} 2\pi R_i L = m L g$$

Upon substitution for the expression of the shear stress

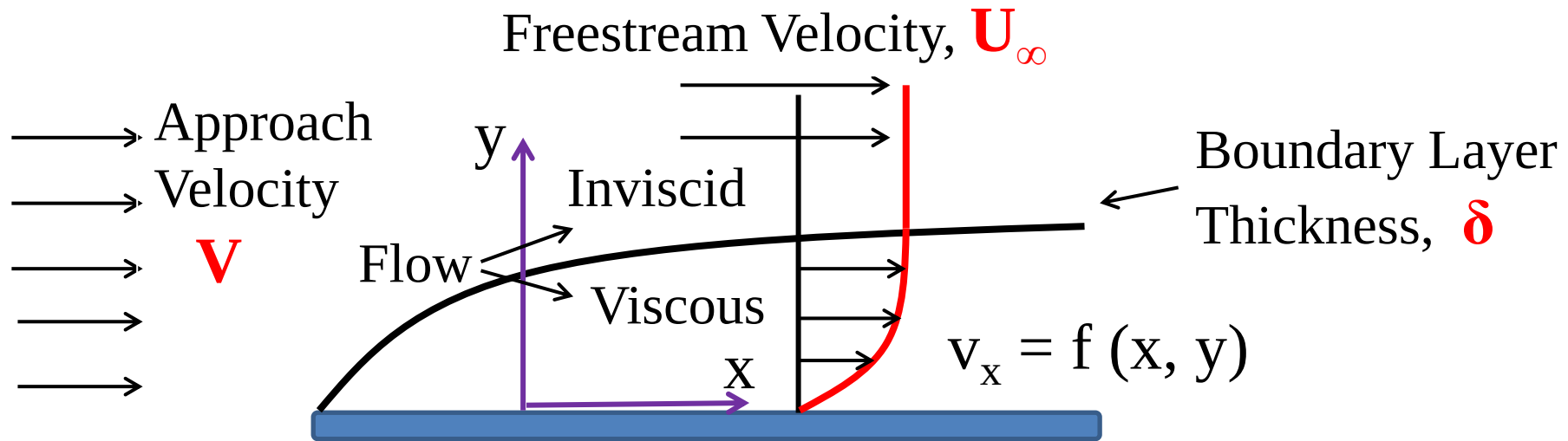
$$V = R_i \ln \frac{R_i}{R_o} \left(\frac{\rho g R_i}{2\mu} - \frac{m g}{2\pi R_i \mu} \right) - \frac{\rho g}{4\mu} (R_i^2 - R_o^2)$$

Order of magnitude analysis of NS Equation

Is it possible to identify the relative magnitudes of the different terms (even approximately)?

It may then be possible to neglect the term(s) that may not play a crucial role in the transport process thereby simplifying NS equations.

External Incompressible Viscous Flow – Boundary Layer



Flow over a flat plate

$v_x = f(x, y)$ Viscous 2D flow inside BL

$v_x = U_\infty$ Inviscid flow outside BL

$v_x = 0.99U_\infty$
at $y = \delta$

δ - boundary layer thickness

Flow Inside the Boundary Layer

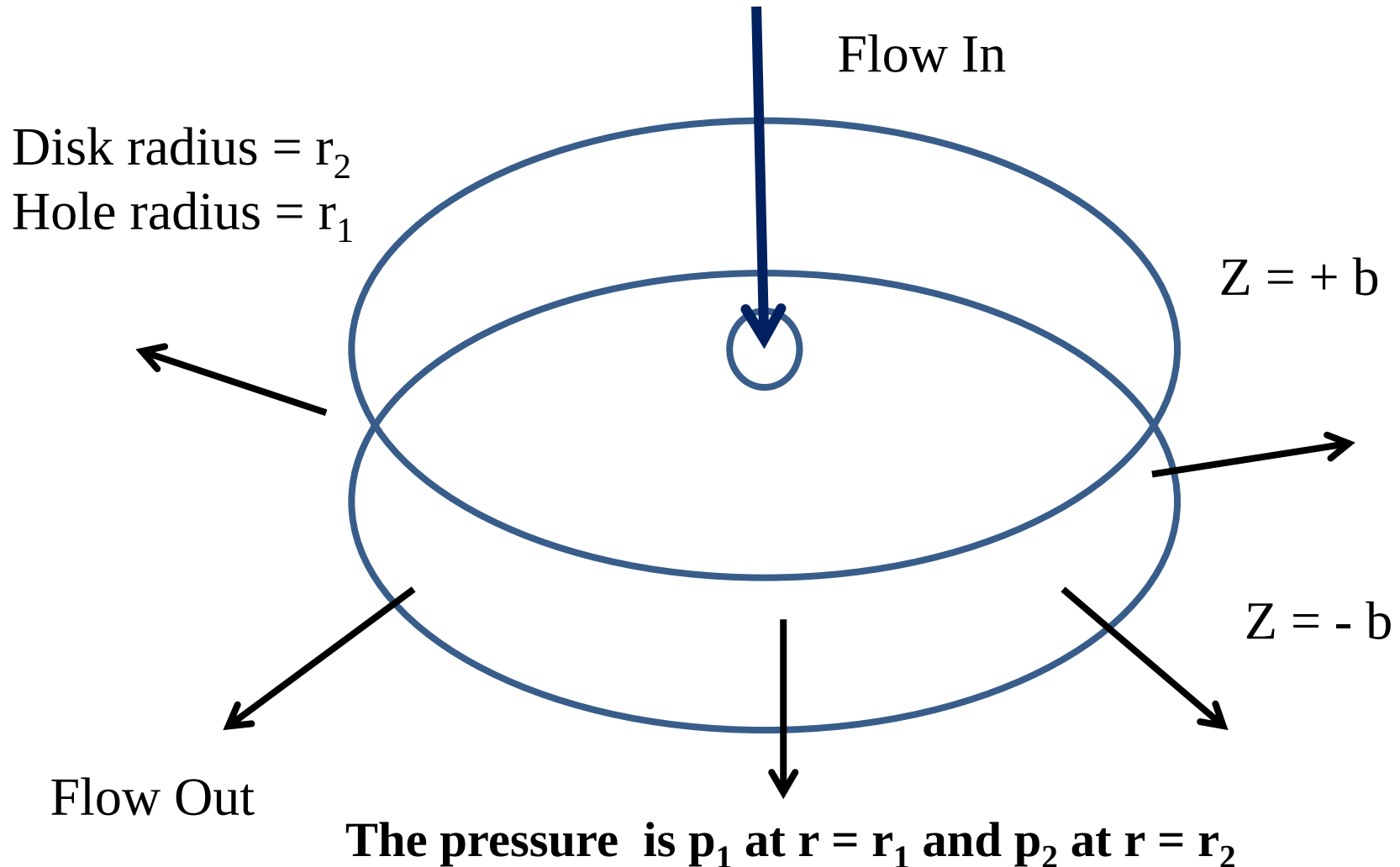
$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \gamma \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right)$$

Boundary Layer Approximations

$$v_x \gg v_y \quad \frac{\partial v_x}{\partial y} \gg \frac{\partial v_x}{\partial x} \quad \frac{\partial^2 v_x}{\partial y^2} \gg \frac{\partial^2 v_x}{\partial x^2}$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \gamma \frac{\partial^2 v_x}{\partial y^2}$$

Flow between two parallel disks with liquid entry through a small hole at the centre of the top plate



$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

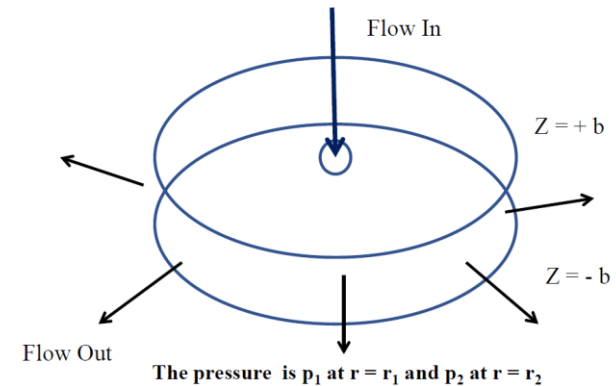
$$\frac{1}{r} \frac{\partial}{\partial r}(r v_r) = 0$$

$$r v_r = \phi \quad v_r$$

$$\Rightarrow v_r = \frac{\phi}{r}, \quad = f(r, z)$$

$$\rho \left(\cancel{\frac{\partial v_r}{\partial t}} + v_r \frac{\partial v_r}{\partial r} + \cancel{\frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta}} - \cancel{\frac{v_\theta^2}{r}} + \cancel{v_z \frac{\partial v_r}{\partial z}} \right)$$

$$= -\frac{\partial P}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2} \right] + \cancel{\rho g_r}$$



$$\phi = f(z)$$

The governing equation

$$\rho \cancel{v_r} \frac{\partial v_r}{\partial r} = - \frac{\partial P}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2} \quad v_r = \frac{\phi(z)}{r}$$

≈ 0

Major assumption on the nature of the flow

CREeping FLOW (CONVECTION
NOT IMP

$$\mu \frac{d^2}{dz^2} (v_r) = - \frac{dP}{dr}$$

$$P = f(r, z)$$

Assume a constant applied pressure difference

$$\frac{\mu}{r} \frac{\partial^2 \phi}{\partial z^2} = \frac{dp}{dr}$$

$$\checkmark \phi = - \frac{\Delta P z^2}{2\mu \ln r_2/r_1} + C_1 z + C_2$$

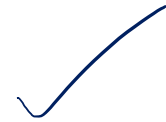
BC $\rightarrow$ NO SLIP $\checkmark$

$$v_r(r, z) = \frac{\phi}{r} = \frac{\Delta P b^2}{2\mu r \ln \frac{r_2}{r_1}} \left[1 - \left(\frac{z}{b} \right)^2 \right]$$

Boundary Conditions

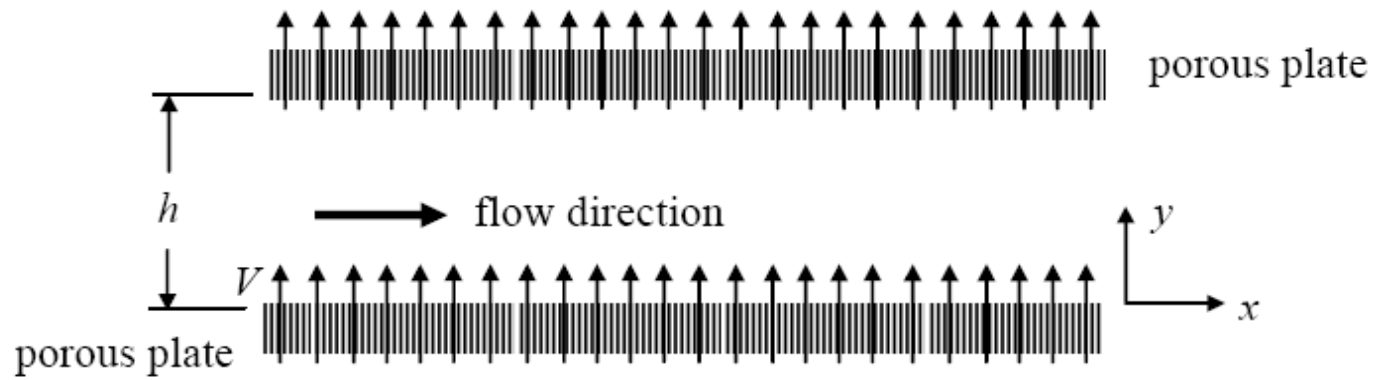


$$v_r(r, z) = \frac{\Delta P b^2}{2 \mu r \ln \frac{r_2}{r_1}} \left[1 - \left(\frac{z}{b} \right)^2 \right]$$



$$Q = 2 \pi \int_{-b}^{+b} r v_r dz = 2 \pi \int_{-b}^{+b} \phi(z) dz = \frac{4 \pi \Delta P b^3}{3 \mu \ln \frac{r_2}{r_1}}$$

An incompressible fluid flows between two porous, parallel flat plates as shown in the figure. An identical fluid is injected at a constant speed V through the bottom plate and simultaneously extracted from the upper plate at the same velocity. Assume the flow to be steady, fully-developed, the pressure gradient in the x -direction is a constant, and neglect body forces.



Determine expressions for the y component of velocity.
Show that the x component of velocity can be expressed as

$$u_x = \frac{h}{\rho V} \left[\frac{\partial p}{\partial x} \right] \left[\left\{ \frac{1 - \exp\left(\frac{\rho V y}{\mu}\right)}{1 - \exp\left(\frac{\rho V h}{\mu}\right)} \right\} - \frac{y}{h} \right]$$

The equation of continuity for fully developed, steady flow in x-direction

$$\cancel{\frac{\partial v_x}{\partial x}} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$= 0 \text{ (Fully dev. flow)}$$

$$\therefore \frac{\partial v_y}{\partial y} = 0 \Rightarrow v = V, \text{ Everywhere in the flow field}$$

The x component of the NS equation

$$\begin{aligned} & \rho \left(\cancel{\frac{\partial v_x}{\partial t}} + v_x \cancel{\frac{\partial v_x}{\partial x}} + v_y \frac{\partial v_x}{\partial y} + \cancel{v_z \frac{\partial v_x}{\partial z}} \right) \\ &= - \frac{\partial P}{\partial x} + \mu \left[\cancel{\frac{\partial^2 v_x}{\partial x^2}} + \frac{\partial^2 v_x}{\partial y^2} + \cancel{\frac{\partial^2 v_x}{\partial z^2}} \right] + \cancel{\rho g_x} \end{aligned}$$

The governing equation

$$\rho v_y \frac{\partial v_x}{\partial y} = - \frac{\partial P}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2}$$

Assumptions:

$\frac{\partial P}{\partial x} = A$

Constant pressure gradient in the x direction ✓

Boundary Conditions

$$v_x = 0 \quad \text{at} \quad y = 0$$

$$v_x = 0 \quad \text{at} \quad y = h$$

$$\mu \frac{d^2 u}{dy^2} = f \vee \frac{du}{dy} - A$$

$\uparrow$
 PR GRAD

$$u = f(\textcircled{0} y)$$

$\frac{BC}{\text{NO SLIP}}$

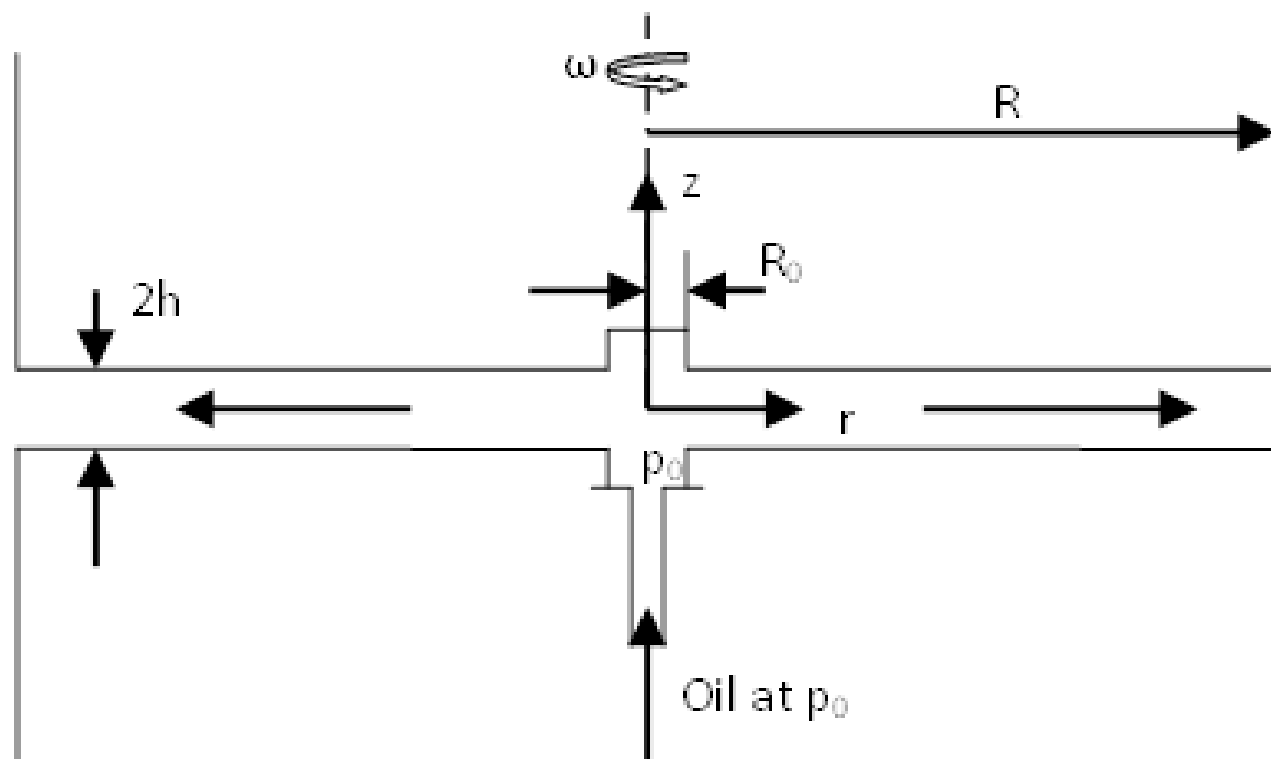
AT $y=0$

$y=h$

Solve the gov. eqⁿ with
the B.C.s to obtain
The profile as provided
in The Question

A thrust bearing as shown in the figure is lubricated by pumping oil at a high pressure of p_0 . The angular velocity is equal to ω . Note that under laminar conditions, both V_r and V_θ in the thin gap will be non-zero and p is a function of r only. Neglect convective and body force terms in equations of motion. You may also assume that the pressure at $r = R$ is equal to p_{atm} , whereas the pressure from $r = 0$ till $r = R_0$ is equal to p_0 .

- (i) Start with the equation of continuity to obtain the functional form of V_r .
- (ii) Show that $V_\theta = \frac{\omega r}{2} \left(1 + \frac{z}{h} \right)$ satisfies the θ component of the NS equation.
- (iii) Write the r component of NS equation to show that $r \frac{\partial p}{\partial r}$ is a constant.
- (iv) Evaluate the pressure distribution using the boundary conditions
- (v) Find the vertical load the bearing can support and the flow rate of oil required.



- (i) Start with the equation of continuity to obtain the functional form of V_r .

The Equation of Continuity

Cylindrical coordinates (r, θ, z) :

$$\cancel{\frac{\partial \rho}{\partial t}} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\cancel{\rho v_\theta}) + \frac{\partial}{\partial z} (\cancel{\rho v_z}) = 0$$

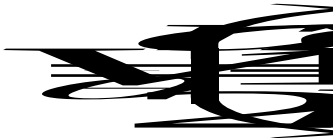
$$\frac{\partial}{\partial r} (r v_r) = 0$$

$$v_r = \frac{f(z)}{r}$$

$$r v_r = f(z)$$

only

$$\frac{\partial}{\partial \theta} (\rho v_\theta)$$

(ii) Show that  satisfies the θ component of the NS equation.

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta$$

$$0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right]$$

$$\frac{\partial^2 v_\theta}{\partial z^2} = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)$$

(iii) Write the r component of NS equation to show that $r \frac{\partial p}{\partial r}$ is a constant

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r$$

$$0 = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r$$

$$0 = -\frac{\partial p}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2}$$

$$r \frac{\partial p}{\partial r} = \mu \frac{\partial^2 (r v_r)}{\partial z^2}$$

$$r \frac{\partial p}{\partial r} = \mu \frac{\partial^2}{\partial z^2} (r v_r)$$

LHS IS A FUNCTION OF r ONLY

RHS IS A FN of z only

$$\therefore r \frac{\partial p}{\partial r} = \mu \frac{\partial^2}{\partial z^2} (r v_r) = \text{CONST.}, K$$

(iv) Evaluate the pressure distribution using the boundary conditions

$$r \frac{\partial p}{\partial r} = K (\text{CONST})$$

$$\text{At } r = r_0 \quad p = p_0, \quad r = R, \quad p = p_{\text{atm}}$$

$$\Rightarrow K = \frac{p_0 - p_{\text{atm}}}{\ln R_0 / R}$$

$$\Rightarrow p = K \ln r + C$$

∴ Non-linear variation of p with r
⇒ p gradient is not const.

$$\cancel{\partial^2 v_r} \frac{\partial^2 (r v_r)}{\partial z^2} = \kappa / \mu$$

$$\text{BC } z = h \quad v_r = 0, \quad z = -h, \\ v_r = 0.$$

$$\Rightarrow v_r = \frac{\kappa}{2\mu r} (z^2 - h^2)$$

$$Q = \int_{-h}^{+h} v_r 2\pi r dz = -\frac{\pi \kappa}{\mu} \frac{4}{3} h^3$$

$$\dot{Q} = \frac{p_0 - p_{atm}}{\ln R/R_0} \frac{4}{3} \frac{\pi h^3}{\mu}$$

Axial load

$$= (p_0 - p_{atm}) \pi R_0^2 + \int_{R_0}^R (p - p_{atm}) 2\pi r dr$$

As

$$p - p_{atm} = k \ln r/R$$

- (v) Find the vertical load the bearing can support and the flow rate of oil required.

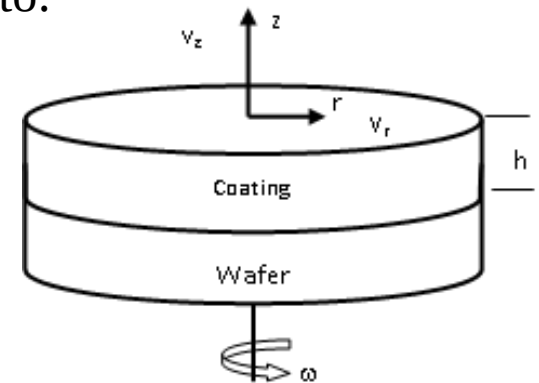
$$\text{Load} = \frac{\pi (p_o - p_{\text{atm}}) (R^2 - R_o^2)}{2 \ln R/R_o}$$

$$Q = \frac{p_o - p_{\text{atm}}}{\ln R/R_o} \frac{4}{3} \frac{\pi h^3}{\mu}$$

Consider the spin-coating process used to coat silicon wafers with photoresist. The process is designed to produce a very thin, uniform coating by spinning a viscous, Newtonian, liquid onto a substrate (wafer). The process has angular symmetry, the rotation rate is constant, and since the film is thin, there are no real pressure gradients or fluid accelerations and body forces to speak of. The thin film also moves with the substrate as if it were a rigid body, $v_\theta \neq f(z)$

a) Show that the continuity and momentum equations reduce to:

$$-\rho \frac{v_\theta^2}{r} = \mu \frac{d^2 v_r}{dz^2} ; \quad \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{\partial v_z}{\partial z} = 0$$



- b) What are the boundary conditions for this problem?
- c) Solve the equations for v_r and v_z .
- d) The velocity, v_z , at the film/air interface is just the change in film thickness with time. Use this to obtain a differential equation for h , and integrate this equation to obtain the solution as:

$$\frac{1}{2} \left(\frac{1}{h^2} + \frac{1}{h_0^2} \right) = \left(\frac{2\rho\omega^2}{3\mu} \right) t$$

Where ω is the rate of rotation ($v_\theta = 2\pi r\omega$) and h_0 is the initial height of the film.

In this case, the following conditions will hold:

$$v_{\theta} = \text{Constant}, g_r = g_{\theta} = g_z = 0, \rho, \mu \text{ constant}$$

$$\frac{\partial}{\partial \theta} = 0, \frac{\partial}{\partial t} = 0, \frac{\partial P}{\partial r} = \frac{\partial P}{\partial \theta} = \frac{\partial P}{\partial z} = 0$$

The continuity and momentum equations in cylindrical coordinates become

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_{\theta}) + \frac{\partial}{\partial z} (\rho v_z) = 0 \quad \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{\partial v_z}{\partial z} = 0$$

Since the fluid moves essentially as a rigid body, $v_r = f(r, z)$. This film is very thin with respect to the radial or angular dimensions so that

$$v_{\theta}, v_r \gg v_z$$

Thus since the angular velocity is constant and v_z is small, we only need to deal with the momentum equation for v_r

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) =$$

$$-\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r$$

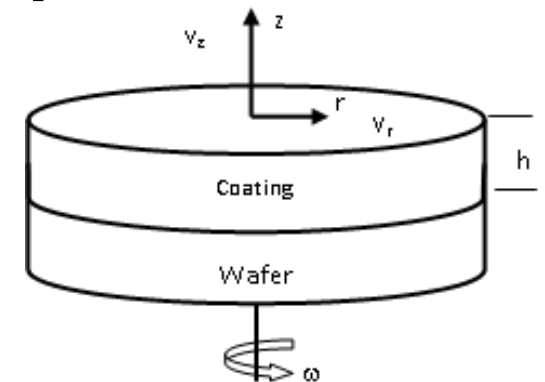
Neglecting all the obvious terms,

$$\rho \left(v_r \frac{\partial v_r}{\partial r} - \frac{v_\theta^2}{r} + v_z \frac{\partial v_r}{\partial z} \right) = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{\partial^2 v_r}{\partial z^2} \right]$$

$$\left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_r}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(r^2 \frac{\partial v_\theta}{\partial \theta} \right) + \frac{\partial^2 v_z}{\partial z^2} \right] = \frac{1}{\eta} \left[\mu \frac{\partial v_r}{\partial r} + \mu \frac{\partial v_\theta}{\partial \theta} + \mu \frac{\partial v_z}{\partial z} \right]$$

Since the film is very thin, v_r changes rapidly with the film thickness, much more rapidly than it changes with the radial position. v_z is quite small as compared to the other components

Thus 1st term on LHS is neglected w.r.t. the second term.



For the same reason, 1st and 3rd terms on the rhs can be neglected.

In one case, v_z is very small and in the second $\frac{\partial v_r}{\partial r}$ is also very small.

v_θ is quite large

Thus

$$\mu \frac{d^2 v_r}{dz^2} + \frac{\rho v_\theta^2}{r} = 0$$

The boundary conditions are

$$z=0; v_r = 0$$

$$z=h; \frac{\partial v_r}{\partial z} = 0$$

The coordinate system is such that $z = 0$ refers to the bottom plate, while $z = h$ is the free surface of the photoresist

Noting that $v_\theta = 2\pi r\omega$, the above equation is solved

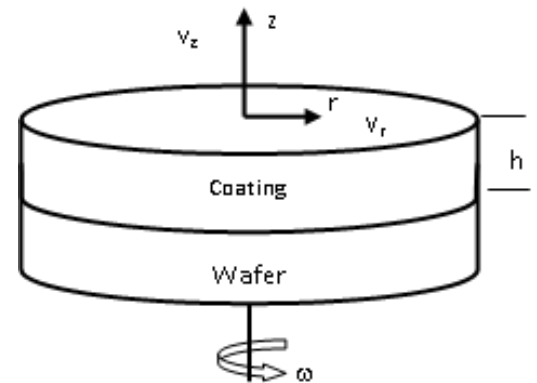
$$v_r = \frac{4 \pi^2 \omega^2}{\gamma} r \left(h z - \frac{z^2}{2} \right)$$

Using the continuity equation

$$\frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{\partial v_z}{\partial z} = 0$$

$$v_z = \frac{4 \pi^2 \omega^2}{\gamma} \left(\frac{z^3}{3} - h z^2 \right)$$

BC: $v_z = 0$ at $z = 0$



$$\langle v_z \rangle = \frac{1}{h} \int_0^h v_z dz = -\frac{\pi^2 \omega^2 h^3}{\gamma} = \frac{dh}{dt}$$

$$\text{At } t=0 \quad h = h_0$$

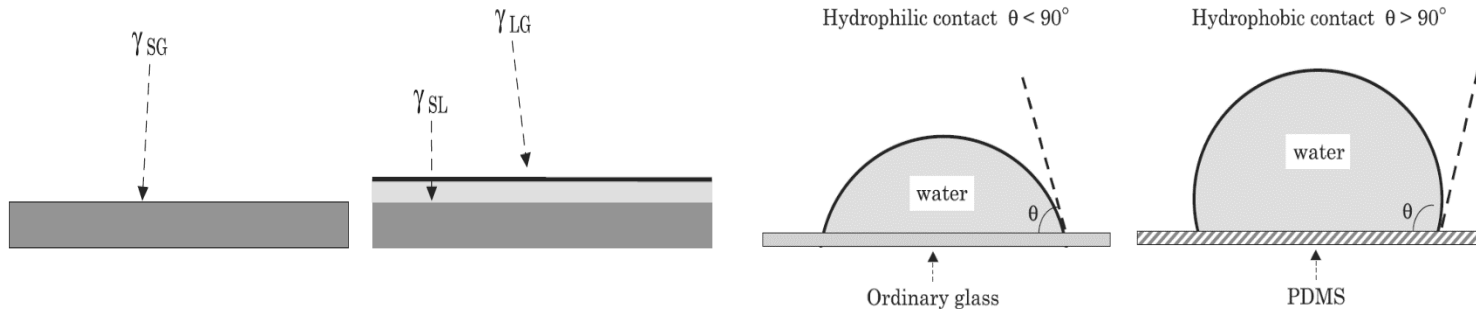
$$\frac{1}{h^2} - \frac{1}{h_0^2} = \frac{2\pi^2 \omega^2}{\gamma} t$$

One of the possible answers

This can also be solved by assuming v_z at $z = h$ to be equal to dh/dt

Supplementary Information – in relation to the query about spreading of a droplet on a solid substrate and contact line motion – slip velocity

Wetting: Partial or Total Wetting



A liquid spreads on a substrate in a film **if the energy of the system is lowered by the presence of the liquid film** .

The surface energy per unit surface of the dry solid surface is γ_{SG} ; the surface energy of the wetted solid is $\gamma_{SL} + \gamma_{LG}$.

The spreading parameter S determines the type of spreading (total or partial)

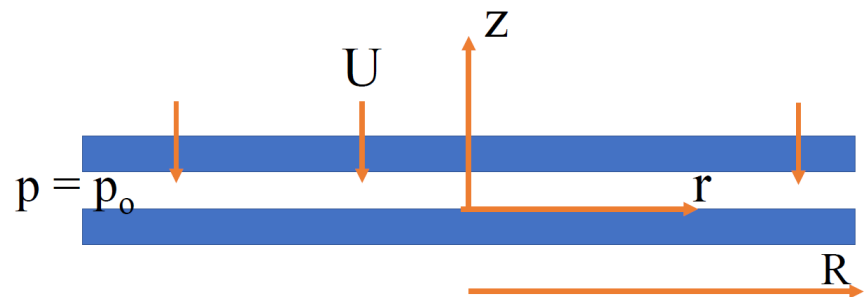
$$S = \gamma_{SG} - (\gamma_{SL} + \gamma_{LG})$$

If $S > 0$, the liquid spreads on the solid surface; if $S < 0$ the liquid forms a droplet.

Problem: One disk approaching another displacing a liquid in between

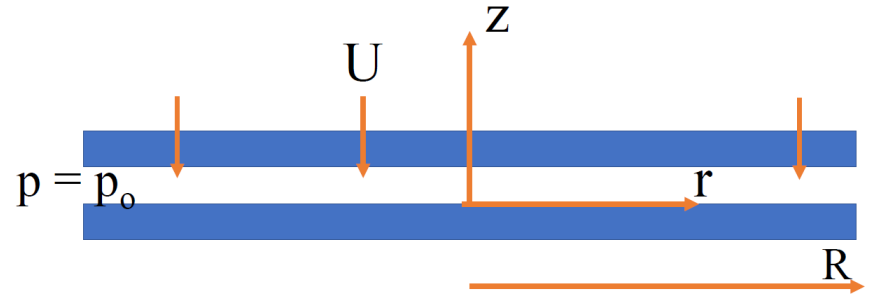
Two parallel, plane circular disks (of radius R) lie one above the other a small distance apart. The space between them is filled with a liquid. The upper disk approaches the lower at a constant (small) velocity U , displacing the liquid. The imposed pressure at $r = R$ is p_o and p_r is not a function of z . Simplify the basic equations (continuity and motion, r component) using an analysis based on your understanding and estimate of the magnitudes of the different terms in the original equations.

Solve these equations with proper boundary conditions to evaluate an expression of the resistance to motion on the moving disk in terms of the separation between the two plates. Take the origin to be at the centre of the lower (stationary) plate.



Assumptions and special considerations

1. R is very large compared to h
2. Leakage rate is small (Small v_r)
3. $\partial p / \partial z$ is negligible
4. v_z is very small as compared to v_r
5. $\partial v_z / \partial z$ may not be small (as z is very small)
6. $\frac{\partial}{\partial z} \left(\frac{\partial v_z}{\partial z} \right)$ can be appreciable
7. v_r is not small as compared to v_z
8. $\frac{\partial v_r}{\partial r}$ is small compared to $\frac{\partial v_r}{\partial z}$
9. $V_\theta = 0$, no θ dependence of v_r , v_z



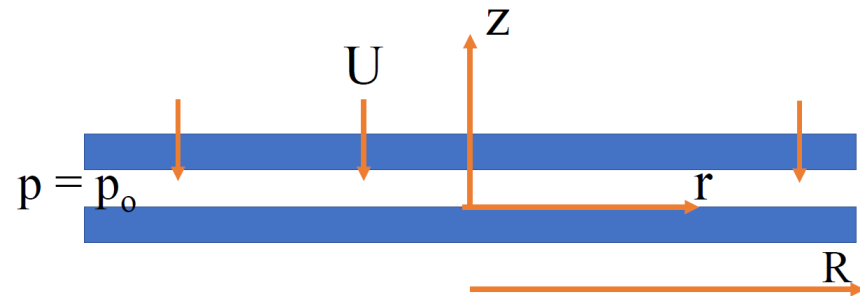
The equation of continuity

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial v_z}{\partial z} = 0 \quad (A)$$

$$\frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{\partial v_z}{\partial z} = 0$$

The terms in the equation

Relative magnitudes of 1,2 w.r.t. 3



The equation of motion (r component)

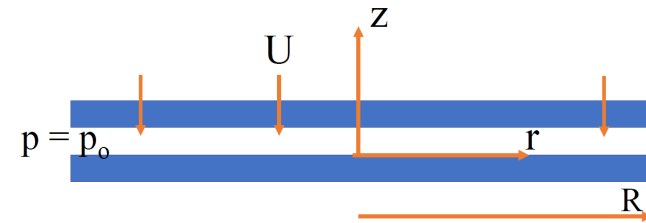
$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) =$$

$$-\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r$$

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial r} + \gamma \left[\frac{1}{r} \frac{\partial v_r}{\partial r} - \frac{v_r}{r^2} + \frac{\partial^2 v_r}{\partial r^2} + \frac{\partial^2 v_r}{\partial z^2} \right]$$

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial r} + \gamma \left[\frac{1}{r} \frac{\partial v_r}{\partial r} - \frac{v_r}{r^2} + \frac{\partial^2 v_r}{\partial r^2} + \frac{\partial^2 v_r}{\partial z^2} \right]$$

Compare terms to arrive at the following equation



$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial r} + \gamma \frac{\partial^2 v_r}{\partial z^2} \quad (B)$$

$$v_r = \frac{1}{2\mu} \frac{\partial P}{\partial r} z^2 + C_1 z + C_2$$

$$BC : \quad z=0; v_r=0 \Rightarrow C_2 = 0$$

$$z=h; v_r=0 \Rightarrow C_1 = -\frac{1}{2\mu} \frac{dp}{dr} h$$

$$v_r = \frac{1}{2\mu} \frac{\partial P}{\partial r} (z - h) z \quad (C)$$

$$\frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{\partial v_z}{\partial z} = 0 \quad (A)$$

Putting the expression of v_r from (C) to (A) and integrating w.r.t. z and as

$$\begin{aligned} BC : \quad z=0; v_r=0 &= v_z \\ z=h; v_r=0; v_z &=-U \end{aligned}$$

$$U = \frac{1}{r} \frac{d}{dr} \int_0^h r v_r dz = \frac{1}{r} \frac{d}{dr} \left(r \frac{dp}{dr} \right) \frac{1}{2\mu} \int_0^h z(z-h) dz = -\frac{h^3}{12\mu} \frac{1}{r} \frac{d}{dr} \left(r \frac{dp}{dr} \right)$$

$$U = -\frac{h^3}{12\mu} \frac{1}{r} \frac{d}{dr} \left(r \frac{dp}{dr} \right)$$

Integrating w.r.t. r

$$-\frac{12\mu}{h^3} \frac{r^2}{2} = r \frac{dp}{dr} + C_1$$

As dp/dr is finite at $r = 0$ $C_1 = 0$ and $p = p_o$ at $r = R$

$$p - p_o = \frac{3\mu U}{h^3} (R^2 - r^2)$$

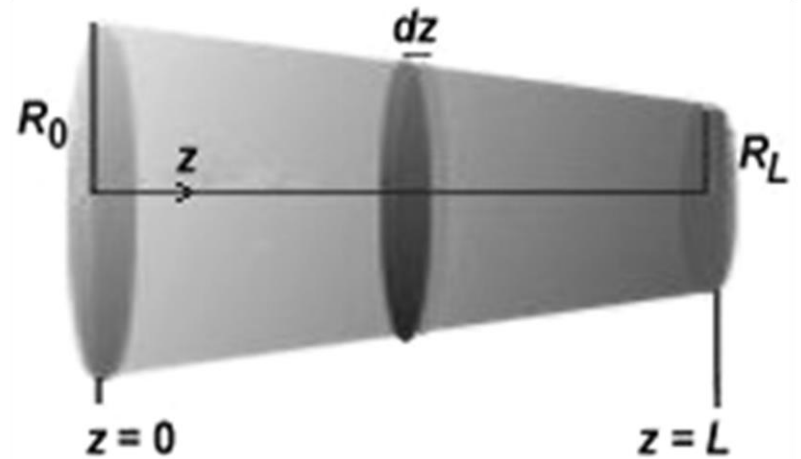
$$\text{Force, } F = \int_0^R 2\pi r dr (p - p_o) = 2\pi \int_0^R \frac{3\mu U}{h^3} (R^2 r - r^3) dr$$

$$\text{Force, } F = \frac{3\pi\mu U R^4}{2h^3}$$

A fluid (of constant density ρ) is in incompressible, laminar flow through a tube of length L . The radius of the tube of circular cross section changes linearly from R_0 at the tube entrance ($z = 0$) to a slightly smaller value R_L at the tube exit ($z = L$).

Using the lubrication approximation, determine the mass flow rate vs. pressure drop (ΔP) relationship for a Newtonian fluid (of constant viscosity μ).

The approximation where a flow between non-parallel surfaces is treated locally as a flow between parallel surfaces is commonly called the lubrication approximation because it is often employed in the theory of lubrication.



The mass flow rate versus the pressure drop (W versus ΔP) relationship for a Newtonian fluid in a circular tube is given by (where $\Delta P = P_0 - P_L$)

$$W = \frac{\pi \Delta P R^4 \rho}{8 \mu L} \quad \text{or} \quad \frac{\Delta P}{L} = \frac{8 \mu W}{\pi \rho R^4} \quad (1)$$

Even for a tapered tube, W does not change with Z

The above equation is assumed to be approximately valid for a differential length dZ of the tube, whose radius R is slowly changing with Z.

Approximation that a flow between non-parallel surfaces can be treated locally as a flow between parallel surfaces is known as LUBRICATION APPROXIMATION.

$$-\frac{dP}{dZ} = \frac{8 \mu W}{\pi \rho} \frac{1}{[R(Z)]^4} \quad (2)$$

$$-\frac{dP}{dZ} = \frac{8\mu W}{\pi \rho} \frac{1}{[R(Z)]^4} \quad (2)$$

$$R(Z) = R_o + (R_L - R_o) \frac{Z}{L}$$

$$\frac{dR}{dZ} = \frac{R_L - R_o}{L} \quad (3)$$

From Eqns 2 and 3

$$-dP = \frac{8\mu W}{\pi \rho} \frac{L}{R_L - R_o} \frac{dR}{R^4}$$

Integrating the above equation between $Z = 0$ ($R = R_o$) and $Z = L$ ($R = R_L$)

$$\int_{P_o}^{P_L} -dP = \frac{8\mu W}{\pi \rho} \frac{L}{R_L - R_o} \int_{R_o}^{R_L} \frac{dR}{R^4}$$

$$\int_{P_o}^{P_L} -dP = \frac{8\mu W}{\pi\rho} \frac{L}{R_L - R_o} \int_{R_o}^{R_L} \frac{dR}{R^4}$$

$$\frac{P_o - P_L}{L} = \frac{8\mu W}{\pi\rho} \frac{L}{3(R_L - R_o)} \left(\frac{1}{R_o^3} - \frac{1}{R_L^3} \right)$$

Thus

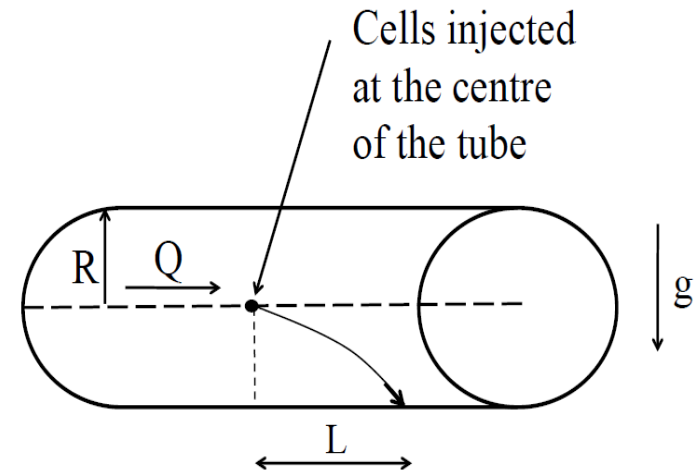
$$W = \frac{\pi \Delta P R_o^4 \rho}{8\mu L} \left[\frac{3(\lambda - 1)}{1 - \lambda^{-3}} \right] ; \quad \lambda = R_L / R_o$$



Taper correction to Hagen-Poiseuille equation

A cell separation (fractionation) system is based on cell density. Cells are injected at the centre of a tube of radius R , and are carried by fluid flowing at a flow rate of Q . Dense cells fall quickly under the action of gravity, adhere to the tube wall and hence do not pass out of the tube. Assume that the concentration of the cell is low enough that the laminar flow in the tube is not perturbed by the presence of the cells. Let the cells be spherical, with radius ' a ' and let them have a density $\rho + \Delta\rho$, where ρ is the density of the flowing fluid.

Find the axial distance (L) that a cell travels before it hits the bottom wall. You need to use the fact that, in fully developed flow in the presence of gravity, the pressure distribution in the vertical distance is hydrostatic and the axial velocity profile is the familiar parabolic shape. You may assume that the cell is spherical, that it reaches its terminal (falling) velocity nearly immediately after injection into the tube, and that it gets carried axially at the local fluid velocity in the tube. State any other assumptions you make.



We need to find the axial distance L that the cell travels before it hits the bottom wall.

The expressions of the forces acting on the spherical cell are:

Gravitational force $F_g = m_{cell} g = \frac{4}{3} \pi a^3 (\rho + \Delta \rho) g$

Buoyancy force $F_b = m_{fluid} g = \frac{4}{3} \pi a^3 \rho g$

Drag force $F_d = 6 \pi \mu a v_r$ The cell is moving downward with a velocity v_r in the r direction

$$\frac{4}{3} \pi a^3 (\rho + \Delta \rho) g = \frac{4}{3} \pi a^3 \rho g + 6 \pi \mu a v_r$$

$$v_r = \frac{2 a^2 g \Delta \rho}{9 \mu}$$

Find the radial location as a function of time as

$$r(t) = \int_0^t v_r dt = \frac{2 a^2 g \Delta \rho t}{9 \mu}$$

The time T, when the cell will hit the wall

$$T = \frac{R}{v_r} = \frac{9 \mu R}{2 a^2 g \Delta \rho}$$

$$v_x = \frac{\Delta P R^2}{4 \mu L} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

$$\langle v_x \rangle = \frac{\Delta P R^2}{8 \mu L}; \quad Q = \pi R^2 \langle v_x \rangle = \frac{\Delta P \pi R^4}{8 \mu L}$$

$$\frac{2Q}{\pi R^2} = \frac{\Delta P R^2}{4 \mu L}$$

$$r(t) = \frac{2a^2 g \Delta \rho t}{9\mu}$$

$$v_x = \frac{2Q}{\pi R^2} \left[1 - \left(\frac{r(t)}{R} \right)^2 \right]$$

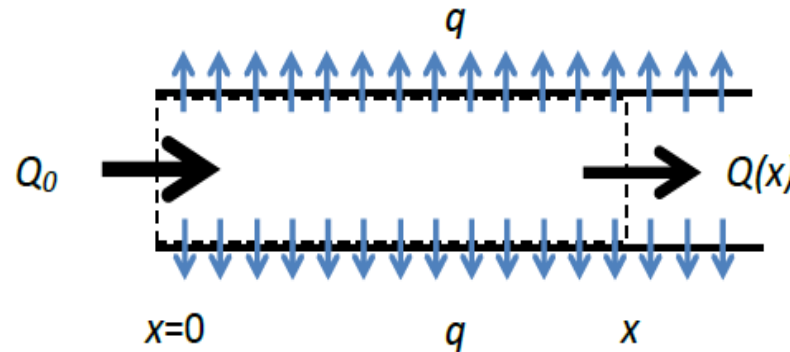
$$L = \int_0^T v_x dt = \frac{2Q}{\pi R^2} \int_0^T \left[1 - \frac{[r(t)]^2}{R^2} \right] dt$$

$$L = \frac{2Q}{\pi R^2} \left[T - \left(\frac{2a^2 g \Delta \rho}{9\mu R} \right)^2 \frac{T^3}{3} \right]$$

$$\text{As } T = \frac{9\mu R}{2a^2 g \rho}$$

$$L = \frac{6\mu Q}{\pi R a^2 g \Delta \rho}$$

Epithelium is one basic type of animal tissue, which lines the cavities and surfaces of structures throughout the body. Epithelial layers contain no blood vessels, so they must receive nourishment via diffusion of substances from the underlying connective tissue, through the basement membrane.



An apparatus has been built for testing the effect of various drugs on the rate at which an epithelium can pump fluid from its luminal side (the side facing the fluid) to its basal side (which lies on the channel wall). The cells line the top and bottom surface of a flow channel that has a separation of h (from top plate to bottom plate; ignore the thickness of cells), a length L , and a depth into the page of W . Each of these walls is porous so that any fluid pumped by the cells can leave the channel. Let each cell layer (top and bottom) pump fluid at a rate of q per unit area of the channel walls (q has thus units of length/time). The height of the channel is much less than its length ($h \ll L$).

Fluid enters the channel at the left at a flow rate Q_0 and a gauge pressure of P_0 . Because of the pumping action of the cells, the flow rate through the channel decreases as a function of x , the distance from the beginning of the channel. To determine the rate at which the cells are pumping fluid out of channel, the channel is instrumented with pressure transducers that can measure $P(x)$. We would like to use this information to find the rate at which the cells pump fluid. The fluid in the channel has a density of ρ and a viscosity of μ . The flow is dominated by viscous effects and is steady.

- (a) Find the pressure distribution, $P(x)$, in the flow channel if $q=0$.
- (b) Find the pressure distribution $P(x)$ in the channel for $q \neq 0$.
- (c) Given that $P(x=L)=P_e$, find q .
- (d) Find the criterion necessary for the assumption that viscous flow dominates to be valid.

All answers must be given in terms of the known quantities e.g., $x, L, W, h, Q_0, P_0, P_e, \rho$ and μ (not all of these parameters need necessarily be used).

(a) Find the pressure distribution, $P(x)$, in the flow channel if $q=0$

For $q = 0$, the situation is simply pressure driven flow between parallel plates.

$$\mu \frac{d^2 v_x}{dy^2} - \frac{dP}{dx} = 0$$

$$\text{At } y = 0, v_x = 0 \Rightarrow C_2 = 0$$

$$v_x = \frac{1}{2\mu} \frac{dP}{dx} y^2 + C_1 y + C_2$$

$$\text{At } y = h, v_x = 0 \Rightarrow C_1 = \frac{1}{2\mu} \frac{dP}{dx} h$$

$$v_x = \frac{1}{2\mu} \frac{dP}{dx} y^2 \left[1 - \frac{h}{y} \right]$$

$$\langle v_x \rangle = \frac{1}{h} \int_0^h v_x dy = \frac{1}{h} \frac{1}{2\mu} \frac{dP}{dx} \left[\frac{y^3}{3} - \frac{h y^2}{2} \right]_0^h = -\frac{1}{12} \frac{dP}{dx} h^2$$

$$Q_0 = W h \langle v_x \rangle = -\frac{1}{12} \frac{dP}{dx} h^3 W$$

$$\frac{dP}{dx} = -\frac{12 \mu}{W h^3} Q_0$$

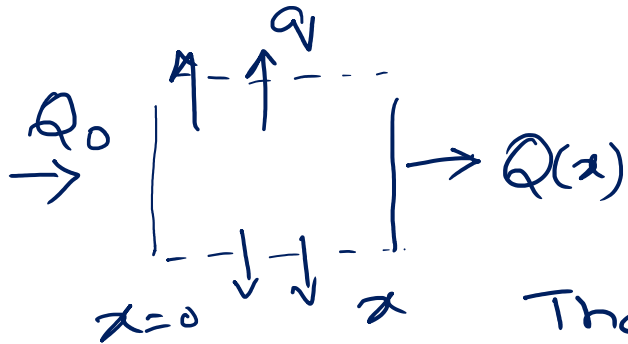
Q_0 is the
flow rate
with $q = 0$

Integrating and imposing the B.C. that $P(x=0) = P_0$

$$P(x) = P_0 - \frac{12 \mu Q_0}{W h^3} x$$

Find the pressure distribution $P(x)$ in the channel for $q \neq 0$.

b)



Mass conservation yields

$$Q(x) = Q_0 - 2qwx$$

The

The assumption of viscous flow allows all inertial terms in the 2D NS Eqn to be neglected

$$\frac{dP}{dx} = \mu \frac{\partial^2 v_x}{\partial x^2} + \mu \frac{\partial^2 v_x}{\partial y^2} \leftarrow \text{Gov. Eqn}$$

So long the trans-cellular velocity at the channel wall is small compared to the axial velocity, i.e., $q \ll Q_0/(Wh)$, the flow Can be treated as nearly Poiseuille Flow and

$$\frac{\partial^2 v_x}{\partial x^2} \ll \frac{\partial^2 v_x}{\partial y^2}$$

There will still be no slip at the channel wall but now with a flow rate that depends on x

$$\frac{dp}{dx} = - \frac{12 \mu Q(x)}{wh^3} = - \frac{12 \mu}{wh^3} [Q_0 - 2q_w x]$$

$$\text{BC } p = p_0 \text{ at } x = 0$$

Solving

$$p = p_0 - \frac{12 \mu x}{wh^3} [Q_0 - q_w x]$$

c)

Given that $P(x=L)=P_e$, find q .

$$P_e = P_0 - \frac{12\mu L}{wh^3} [Q_0 - qwL]$$

∴

$$q = \frac{Q_0}{wL} - \frac{P_0 - P_e}{12\mu L^2} h^3$$

(d) Find the criterion necessary for the assumption that viscous flow dominates to be valid.

The inertial terms must be quite small as compared to the viscous terms

$$\frac{\rho \left(v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} \right)}{\mu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right)} \ll 1$$

Using the result from mass conservation in (b)

$$\frac{\partial v_x}{\partial x} \sim \frac{\partial}{\partial x} (Q/A) = \frac{d}{dx} \left(\frac{Q}{wn} \right) = \frac{1}{wn} \frac{d}{dx} (Q_0 - \frac{2Q_0 \sqrt{wx}}{2})$$

$$= \frac{2q}{n} \sim \frac{q}{n}$$

$$v_y \sim q, \quad \frac{\partial v_x}{\partial y} \sim \frac{v_x}{h}, \quad \frac{\partial^2 v_x}{\partial y^2} \ll \frac{\partial^2 v_x}{\partial x^2}$$

$$\frac{\partial^2 v_x}{\partial y^2} \sim \frac{\cancel{v_x}}{\cancel{h^2}} \frac{v_x}{h^2}$$

$$\therefore \frac{\rho \left(v_x \frac{q}{h} + q \frac{v_x}{h} \right)}{\mu v_x / h^2} \ll 1, \Rightarrow \frac{\rho q h}{\mu} \ll 1$$

$\frac{\rho q h}{\mu}$ = Effective Re no., but depends on leakage flow

The Equation of Continuity

$$[\partial\rho/\partial t + (\nabla \cdot \rho\mathbf{v}) = 0]$$

Cartesian coordinates (x, y, z):

$$\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0 \quad (\text{B.4-1})$$

Cylindrical coordinates (r, θ , z):

$$\frac{\partial\rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0 \quad (\text{B.4-2})$$

Spherical coordinates (r, θ , ϕ):

$$\frac{\partial\rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r}(\rho r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(\rho v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}(\rho v_\phi) = 0 \quad (\text{B.4-3})$$

^a When the fluid is assumed to have constant mass density ρ , the equation simplifies to $(\nabla \cdot \mathbf{v}) = 0$.

The Equation of Motion for a Newtonian Fluid with constant μ , ρ

The Navier Stokes Equation

Cartesian coordinates (x, y, z):

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left[\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right] + \rho g_x$$

$$\rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left[\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right] + \rho g_y$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

Cylindrical coordinates (r, θ, z) :

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) &= -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r \\ \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta \\ \rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \end{aligned}$$

Spherical coordinates (r, θ, ϕ) :

$$\begin{aligned} \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\theta^2 + v_\phi^2}{r} \right) &= -\frac{\partial p}{\partial r} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial^2}{\partial r^2} (r^2 v_r) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_r}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_r}{\partial \phi^2} \right] + \rho g_r \\ \rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} + \frac{v_r v_\theta - v_\phi^2 \cot \theta}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\theta}{\partial \phi^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} - \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right] + \rho g_\theta \\ \rho \left(\frac{\partial v_\phi}{\partial t} + v_r \frac{\partial v_\phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\phi}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi v_r + v_\theta v_\phi \cot \theta}{r} \right) &= -\frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} \\ &+ \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\phi}{\partial \phi^2} + \frac{2}{r^2 \sin \theta} \frac{\partial v_r}{\partial \phi} + \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right] + \rho g_\phi \end{aligned}$$